ENGR 290 Multiple Choice Practice

60 questions • No calculator • Choose one answer per question

Use a separate answer sheet. Questions 1–25 cover crystals, defects, slip and diffusion. Questions 26–60 extend through solidification, polymers, phase diagrams and mixed review. Explanations follow Question 60.

1 How many atoms belong to one conventional FCC unit cell?

A. 8
B. 4
C. 1
D. 2

2 Which relation connects atomic radius r and BCC cell edge a?

A. $a = 4r$
B. $\sqrt{2}a = 4r$
C. $\sqrt{3}a = 4r$
D. $a = 2r$

3 A cube has side 20.0 mm and mass 32.0 g. Its density is

A. 0.004 g/cm^3
B. 0.8 g/cm^3
C. 4.00 g/cm^3
D. 1.6 g/cm^3

4 What distinguishes a crystalline solid from an amorphous solid?

A. No grain boundaries
B. No defects
C. Only ionic bonds
D. Long-range periodic atomic order

5 An atom on a unit-cell face contributes what fraction to that cell?

A. 1
B. 1/2
C. 1/8
D. 1/4

Questions 6 to 10

6 Which notation represents a single crystal direction?

A. ⟨110⟩
B. (110)
C. [110]
D. {110}

7 Intercepts a, 2b, and ∞ give which Miller indices?

A. (201)
B. (120)
C. (210)
D. (110)

8 The conventional intercepts of (311) are

A. a/3, b, c
B. a/3, b/3, c/3
C. a, b/3, c
D. 3a, b, c

9 For a cubic crystal with a = 0.360 nm, d_{200} equals

A. 0.180 nm
B. 0.255 nm
C. 0.360 nm
D. 0.720 nm

10 A zero third Miller index means a plane is

A. Normal to the z axis
B. Parallel to the z axis
C. Not a valid plane
D. Through the origin

Questions 11 to 15

11 Which bond type uses delocalized electrons?

A. Covalent
B. Ionic
C. Metallic
D. Hydrogen

12 An ion leaves its site and occupies an interstitial position. This creates

A. A grain boundary
B. A screw dislocation
C. A Schottky pair
D. A Frenkel defect

13 Which temperature should enter $\exp(-E_v/kT)$ with k in eV/K at 400 °C?

A. 127
B. 1073.15
C. 400
D. 673.15

14 For $f = C\exp(-E_v/kT)$, the slope of ln f versus 1/T is

A. $-k/E_v$
B. C
C. E_v/k
D. $-E_v/k$

15 What happens to equilibrium vacancy fraction as temperature increases?

A. It decreases
B. It increases
C. It stays constant
D. It always vanishes

Questions 16 to 20

16 Which is a substitutional defect?

A. A boundary between grains
B. A host atom in a gap
C. An empty lattice site
D. A solute atom replacing a host atom

17 For a pure edge dislocation, the Burgers vector is

A. Always zero
B. Normal to every slip plane
C. Parallel to its line
D. Perpendicular to its line

18 With $\sigma = 100$ MPa and $\varphi = \lambda = 45°$, the resolved shear stress is

A. 200 MPa
B. 25 MPa
C. 50 MPa
D. 100 MPa

19 Reducing grain diameter to one quarter makes the Hall–Petch grain-size contribution

A. Unchanged
B. Four times as large
C. Twice as large
D. Half as large

20 Why can cold working increase strength?

A. It makes bonds ionic
B. More dislocations obstruct one another
C. It eliminates all grains
D. It removes every dislocation

Questions 21 to 25

21 The minus sign in Fick's first law means flux goes

A. Only in the negative x direction
B. Only toward hotter regions
C. Up the concentration gradient
D. Down the concentration gradient

22 For C_0 = 1.0, C = 0.2, C = 0.6 wt %, (C−C

A. 0.2
B. 0.5
C. 0.6
D. 0.4

23 If normalized concentration is 0.3 in the semi-infinite solution, erf(z) is

A. −0.3
B. 0.7
C. 1.3
D. 0.3

24 At fixed D and target concentration, doubling depth changes diffusion time by

A. ×1/4
B. ×1/2
C. ×4
D. ×2

25 Which assumption is used in the error-function carburizing solution here?

A. Constant surface concentration and constant D
B. No concentration gradient
C. A finite slab already uniform everywhere
D. D must be zero

Questions 26 to 30

26 A dendrite tip grows out of a flat front into hotter liquid. What happens next?

A. It becomes a grain boundary
B. Its latent heat vanishes
C. It accelerates indefinitely
D. It stalls or remelts

27 Why can seeding refine a casting?

A. The equilibrium freezing point doubles
B. Every grain grows faster
C. All nuclei dissolve
D. More grains start growing

28 Which term opposes forming a very small solid nucleus?

A. Latent heat removal alone
B. New interfacial energy
C. Loss of liquid volume alone
D. Bulk free-energy decrease

29 Which is a benefit of a single-crystal blade?

A. Eliminating all slip
B. Removing all point defects
C. Making the metal amorphous
D. Eliminating grain-boundary sliding

30 What produces a thermal arrest in a pure metal?

A. Zero heat transfer
B. Release of latent heat
C. Creation of two components
D. Vaporization of all solid

Questions 31 to 35

31 Which polymer would recover best after a large extension?

A. Flexible chains with sparse cross-links
B. Unconnected short molecules
C. A glassy highly filled resin
D. A dense rigid network

32 What changes at Tg in an amorphous thermoplastic on warming?

A. Primary bonds all break
B. Segmental mobility increases
C. Molar mass doubles
D. All crystallites melt

33 A sample is drawn and retested along the draw axis. Why can it be stronger?

A. Its chains align with the load
B. All cross-links dissolve
C. Its repeat units become heavier
D. It gains metallic bonds

34 Which process suits a continuous cable jacket?

A. Extrusion
B. Casting a single cavity
C. Batch injection molding
D. Crucible seeding

35 Which observation identifies stress relaxation?

A. Stress falls at fixed strain
B. Stress rises at fixed strain rate
C. Strain rises at fixed stress
D. Mass falls at fixed volume

Questions 36 to 40

36 At equal repeat mass, chain A has twice the molar mass of B. Its DP is…

A. Twice as large
B. Unchanged
C. Half as large
D. Four times as large

37 Which change generally improves impact resistance in ABS?

A. More rubbery BS fraction
B. Prevent energy absorption
C. More rigid SAN fraction
D. Remove the rubber phase

38 Which reaction often releases a small molecule as chains grow?

A. Addition of ethylene
B. Vulcanization by definition
C. Physical drawing
D. Condensation

39 Which candidate is most likely to remain rigid on reheating after cure?

A. Densely cross-linked thermoset
B. Softened linear polyethylene
C. Oligomer melt
D. Uncross-linked thermoplastic

40 On slow cooling a liquid alloy, where does the first solid appear?

A. Solidus
B. Glass transition
C. Solvus
D. Liquidus

Questions 41 to 45

41 How do you read the composition of the first solid?

A. Use the liquidus point only
B. Read the opposite lever-arm length as wt.% solute
C. Use the overall composition always
D. Follow the tie line to its solid boundary

42 A tie line runs from 10 to 50 wt.% B. At 20 wt.% B the left phase is…

A. One quarter of the mass
B. Absent
C. Three quarters of the mass
D. Half the mass

43 Which reading gives the freezing range?

A. Liquidus minus solidus temperature
B. Temperature at the triple point
C. Eutectic composition minus solubility
D. Distance across a tie line

44 What transforms at the eutectic temperature in a hypoeutectic alloy?

A. Remaining liquid
B. Only primary α
C. The entire alloy if primary α is present
D. Only gas

45 Which reaction is peritectic on cooling?

A. Solid → two solids
B. Two solids → liquid
C. Liquid → two solids
D. Liquid + solid → new solid

Questions 46 to 50

46 A single solid becomes two new solids on cooling. Name the reaction.

A. Peritectic
B. Monotectic
C. Eutectoid
D. Eutectic

47 Which reaction has two distinct liquid phases in its equation?

A. Monotectic
B. Eutectic
C. Eutectoid
D. Peritectoid

48 What forms first on slow cooling 0.4 wt.% C steel from austenite?

A. Proeutectoid ferrite
B. Proeutectoid cementite
C. Liquid
D. Pure graphite by the lecture diagram

49 Which description fits pearlite?

A. A single FCC phase
B. Ferrite and cementite in layers
C. Two liquids of different composition
D. Pure primary ferrite

50 In a binary three-phase field at invariant temperature and fixed pressure, F equals…

A. One
B. Three
C. Two
D. Zero

Questions 51 to 55

51 Why does adding 10 kg solute to 100 kg alloy change the denominator?

A. Composition is measured in moles only
B. The final mass is 110 kg
C. The original solute disappears
D. Only the initial solvent counts

52 Which boundary gives the solubility limit within solid material?

A. Solvus
B. Liquidus
C. Critical-radius curve
D. Vapour-pressure curve

53 Two differently oriented grains of chemically identical FCC copper are…

A. Necessarily two phases
B. Two grains of one phase
C. One grain of two phases
D. An amorphous phase

54 In a solid solution, what distinguishes an interstitial solute?

A. It removes equal numbers of ions
B. It replaces a host atom
C. It sits in a gap between normal sites
D. It terminates a half-plane

55 Which combination favours extensive substitutional solubility?

A. Large size mismatch and different structures
B. Only a shared colour
C. Similar size and crystal structure
D. A fixed compound formula

Questions 56 to 60

56 What makes a phase metastable?

A. It has no crystal structure
B. A kinetic barrier delays a lower-energy state
C. It has the lowest possible energy
D. Its composition must be pure

57 What does an ideal screw dislocation permit that a pure edge segment cannot do by simple glide?

A. Remove every vacancy
B. Cross-slip onto another compatible plane
C. Glide without a Burgers vector
D. Create a free surface every step

58 At the triple point of a pure substance, varying temperature independently while retaining all phases is…

A. Possible only for metals
B. Always possible at any pressure
C. Possible if the sample mass doubles
D. Impossible at equilibrium

59 A cured elastomer is vulcanized more heavily. Which structural change occurs?

A. All chains become linear
B. More links between chains
C. Fewer backbone atoms per repeat
D. All secondary bonds disappear

60 If the same mass occupies half the volume, the density is…

A. Unchanged
B. Twice as large
C. One quarter as large
D. Half as large

Answer key 1 to 12

1 B 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

2 C Atoms touch along the BCC body diagonal, whose length is $\sqrt{3}a$.

3 C 20.0 mm = 2.00 cm, so V = 8.00 cm^3 and ρ = 32.0/8.00 = 4.00 g/cm^3.

4 D Crystalline solids have long-range periodic order. Real crystals can contain defects.

5 B A face atom is shared between two neighbouring cells.

6 C Square brackets indicate a direction; angle brackets denote a family of directions.

7 C Reciprocals are 1, 1/2, 0; multiplying by 2 gives (210).

8 A Take reciprocal indices: a/3, b/1, c/1.

9 A $d_{200} = a/\sqrt{(2^2+0^2+0^2)} = a/2$.

10 B Zero is the reciprocal of an infinite intercept, indicating parallelism to that axis.

11 C Metallic bonding has mobile, delocalized electrons rather than localized electron pairs.

12 D The associated vacancy and same-ion interstitial form a Frenkel defect.

Answer key 13 to 24

13 D $T(K) = T(°C) + 273.15 = 673.15$ K.

14 D $\ln f = \ln C - (E_v/k)(1/T)$, so the slope is negative.

15 B The negative exponent becomes less negative, increasing the equilibrium fraction.

16 D A substitutional atom occupies a regular lattice site that would otherwise hold a host atom.

17 D An edge dislocation has b perpendicular to its line; a screw dislocation has b parallel to its line.

18 C $\tau = 100 \cos 45° \cos 45° = 50$ MPa.

19 C The contribution is proportional to $1/\sqrt{d}$; $\sqrt{4} = 2$. The baseline σ_0 is unchanged.

20 B The increased dislocation density creates interactions that hinder subsequent motion.

21 D Flux is opposite to the concentration gradient; its coordinate sign depends on that gradient.

22 B $(0.6-0.2)/(1.0-0.2) = 0.4/0.8 = 0.5$.

23 B The normalized concentration is $1 - \text{erf}(z)$, so $\text{erf}(z) = 0.7$.

24 C Rearranging $z = x/(2\sqrt{Dt})$ gives $t = x^2/(4Dz^2)$.

Answer key 25 to 36

25 A The solution assumes a semi-infinite body, uniform initial concentration, constant surface concentration, and constant D.

26 D The hotter liquid suppresses freezing at the tip.

27 D More nuclei share the available material before their grains impinge.

28 B Creating a new solid–liquid interface costs energy proportional to area.

29 D One continuous grain has no internal grain boundaries.

30 B Energy leaves while the liquid transforms at the freezing temperature.

31 A Flexible segments uncoil and cross-links restore the network.

32 B Cooperative segment motion becomes possible and modulus falls.

33 A Alignment transfers more axial load to covalent backbones.

34 A A die shapes a continuous length with a fixed cross-section.

35 A A held deformation needs progressively less stress as chains rearrange.

36 A Degree of polymerization is chain molar mass divided by repeat molar mass.

Answer key 37 to 48

37 A Rubber domains dissipate energy and can improve impact toughness.

38 D Condensation commonly eliminates water or another small molecule.

39 A A permanent network prevents chains from flowing past one another.

40 D The liquidus separates the all-liquid field from liquid plus solid.

41 D The solid endpoint gives its composition. Lever arms give phase amounts.

42 C Its opposite arm spans 30 of the total 40 percentage points.

43 A The interval runs from first solid to last liquid disappearing.

44 A Residual liquid transforms into two solids while primary α persists.

45 D A liquid reacts with a pre-existing solid at the invariant temperature.

46 C All phases in a eutectoid reaction are solid.

47 A One liquid produces a second liquid and a solid.

48 A It lies below the lecture eutectoid composition of 0.8 wt.% C.

Answer key 49 to 60

49 B Pearlite is the two-phase product of the eutectoid transformation.

50 D $F = C - P + 1 = 2 - 3 + 1$.

51 B Final mass fraction uses final solute mass over final total mass.

52 A Crossing a solvus can lead to precipitation of another solid phase.

53 B Orientation distinguishes grains while composition and crystal structure identify this phase.

54 C Small solutes can occupy spaces in the host lattice.

55 C Similar radii and structure reduce mismatch. Valence and electronegativity also matter.

56 B A phase can persist in a local energy minimum.

57 B A screw line and its parallel Burgers vector can lie in multiple slip planes.

58 D The full phase rule gives $1 - 3 + 2 = 0$.

59 B Sulfur bridges connect neighbouring chains and constrain motion.

60 B Density equals mass divided by volume.